PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Malnutrition assessment methods in adult patients with tuberculosis: a systematic review
AUTHORS	ter Beek, Lies; Bolhuis, Mathieu; Jager-Wittenaar, Harriët; Brijan, Rene; Sturkenboom, Marieke; Kerstjens, Huib; de Lange, Wiel; Tiberi, Simon; van der Werf, Tjip S.; Alffenaar, Jan-Willem; Akkerman, Onno W.

VERSION 1 – REVIEW

REVIEWER	Cervantes, Jorge Texas Technical University Health Sciences Center - El Paso Campus
REVIEW RETURNED	20-Mar-2021

GENERAL COMMENTS	I have reviewed the paper by Ter Beek et al. The statement “malnutrition is the most important risk factor for re-activation of TB,” appears rather bold, and not only a more recent reference, but a more comprehensive one is needed. Since they state that the WHO released guidelines for nutrition assessment in TB patients, the aim of the Review should not be to identify malnutrition assessment methods, but to see how much it has been implemented. The later cannot be done just through a systematic literature review. I suggest to change the title to a more honest one, realizing the limitations of the approach. The section Data analysis, doesn’t present any statistical analysis. It is just purely descriptive. Even a size effect would have been appreciated.

REVIEWER	Cox, Sharon London School of Hygiene & Tropical Medicine, MRC International Nutrition Group
REVIEW RETURNED	07-Apr-2021

GENERAL COMMENTS	General comments. The manuscript addresses the important issue of a lack of consistency and understanding of how malnutrition can be assessed in Tuberculosis. Arguably, this results from, in part, the complexity of a condition with such a strong bi-directional association with malnutrition and the vertical and silo nature of TB programmes. However, I don’t feel that the manuscript fully addresses this aspect of the issue. The discussion also does not address the limitations of the current 2013 WHO guidance document in that it does not describe or refer to methods of assessment of malnutrition. As this manuscript is written for a

	general audience in this journal, I suggest that more explanation of some of the terms and background description is included of things like the malnutrition screening tools is included – perhaps as supplementary material. The manuscript describes a systematic review of published papers in which malnutrition was assessed within populations of adult persons with (presumed or microbiologically confirmed) active TB disease and describes and evaluates the methods used for malnutrition assessment. The manuscript then attempts to determine if methods changed after the publication of the WHO guidelines on the management of malnutrition in TB in 2013. The stated aims, objectives, described methods and presented results are not fully consistent with each other and I suggest that the wording of these should be carefully reviewed and edited for clarity, precision and consistency. For example, the title and the primary aim at the end of the introduction state that the purpose is to describe current practice in TB, but the review is only of published research papers, which is unlikely to reflect general practice outside of research studies. I suggest changing the stated objectives to reflect this and/or including in the discussion. The secondary aim/objective is to “quantify the malnutrition assessments that reflect the international consensus definition for malnutrition in this population” [intro, line 112-13] or “cover the three domains of the construct of malnutrition according to its international consensus definition” [abstract]. However, as stated by the authors [line 252, discussion] “the GLIM criteria do not include the domain of functionality as a criteria [sic]”. Indeed, in the main, most recent GLIM document (Cederholm, Clinical Nutrition 2019) the document does not refer to these “domains” at all. Instead, the definitions are around definitions and diagnoses for phenotypic and aetiologic criteria for defining malnutrition. Further in the ESPEN guidelines on the definitions and terminology of clinical nutrition (ref 14) this uses these criteria to define into categories and sub-categories of disease-related malnutrition and malnutrition without disease, and these terms are not discussed in the current manuscript. I feel there is a real problem and confusion in the manuscript in trying to include physical and cognitive as both a conceptual domain of malnutrition and a diagnostic criterion. In disease-related malnutrition it is impossible to separate out the impacts and aetiology of these from the disease or malnutrition. Also, how do measures like grip strength (which was not included in the reported measures of malnutrition, but is included in some of the included manuscripts) fit into these domains. In the GLIM documents, grip-strength is included as a proxy for measuring body composition of reduced muscle mass (domain B), but yet is also clearly functional (domain C?). The MNA that is include as an assessment tool for malnutrition is included as method that includes all three domains, but yet it is really a malnutrition screening tool and I question that it really addresses the “physical and cognitive domain” that people not familiar with the tool might conceive. For the reporting of the results and characteristics of the included studies I suggest: (1) reporting and separating out studies based on why they assessed malnutrition, was it assessed as a primary outcome of the study, or as a main exposure or for other purposes

	and if so what? E.g. for determining those who were to receive some kind of intervention – nutritional or otherwise? (2) Related to this, reporting if participants in studies were inpatients or outpatients; (3) reporting and analysing studies by year of data collection rather than publication. In recent studies, this should always be reported. Specific/minor comments Abstract: The objectives don't match the title of the manuscript. The methods section needs more details of the search criteria. I suggest the whole abstract needs re-writing as per PRISMA guidelines and above comments Introduction: The statement "70% of TB are malnourished needs referencing of the original source, not the Bhargava paper which concerns only a particular population in India. Methods: The methods state that the papers selected that "aimed to assess malnutrition and described a method for assessment of malnutrition in patients with microbiologically confirmed TB". (1) I know many of the studies included clinically diagnosed TB.

VERSION 1 – AUTHOR RESPONSE

Comments from Reviewer 1:

- Comment 1: The statement "malnutrition is the most important risk factor for re-activation of TB," appears rather bold, and not only a more recent reference, but a more comprehensive one is needed.

- Author reply: Thank you for this suggestion, we have changed the sentence into is 'an important risk factor' and added an extra reference (WHO Global Tuberculosis report 2020) (line 106-108).

- Revised sentence: As hunger-related malnutrition caused by food-insecurity impacts the immune system, malnutrition is an important risk factor for re-activation of TB, with a reported 27% attributable risk.

- Comment 2: Since they state that the WHO released guidelines for nutrition assessment in TB patients, the aim of the Review should not be to identify malnutrition assessment methods, but to see how much it has been implemented. The later cannot be done just through a systematic literature review. I suggest to change the title to a more honest one, realizing the limitations of the approach.

- Author reply: Thank you for your comment. Our intention was to review nutritional assessment methods and see how they capture the conceptual definition of malnutrition. To improve clarity on this objective we removedhe sentence on implementation of the WHO guideline on nutritional care from the manuscript (line 134-135) and revised the title.

- Revised title: Malnutrition assessment methods in adult patients with tuberculosis: a systematic review

- Comment 3: The section Data analysis, doesn't present any statistical analysis. It is just purely descriptive. Even a size effect would have been appreciated.

- Author reply: This is a correct statement, however statistical analysis are not applicable for our objectives. In addition, we cannot add new statistic tests after designing and performing the study.

Comments from Reviewer 2:

- Comment 1: The manuscript addresses the important issue of a lack of consistency and understanding of how malnutrition can be assessed in Tuberculosis. Arguably, this results from, in part, the complexity of a condition with such a strong bi-directional association with malnutrition and the vertical and silo nature of TB programmes. However, I don't feel that the manuscript fully addresses this aspect of the issue. The discussion also does not address the limitations of the current 2013 WHO guidance document in that it does not describe or refer to methods of assessment of malnutrition.

- Author reply: We very much agree, these are two very relevant observations (the WHO guideline refers only to BMI as an assessment method) and we have added these to both the introduction and the discussion (line 130-134 and line 278-279).

- Revised section of the introduction: It was not until 2013 that the World Health Organization (WHO) presented their first guideline on nutritional care and support for TB patients. In this guideline, the WHO stressed that all patients with active TB receive individualized nutritional assessment and management, including dietary counselling and nutritional interventions, to improve nutritional status and consequently, prevent TB treatment failure. Nutritional assessment is a necessary step in the 'nutritional care process', enabling the professional to design a treatment plan together with the patient. However, the 2013 WHO guideline only refers to BMI as a method of assessing malnutrition for adults. The important issue of lack of consistency and understanding of how malnutrition can be assessed in patients with TB results from, the complexity of TB being on the one hand strongly associated with malnutrition and on the other by the vertical and siloed nature of TB programmes.

- Revised section in the discussion: This review indicates that the international criteria for the assessment of malnutrition have not yet found their way into studies with TB patients. The 2013 WHO guideline on nutritional care only refers to BMI as a method of assessing malnutrition for adults, which may contribute to the status quo.

- Comment 2: As this manuscript is written for a general audience in this journal, I suggest that more explanation of some of the terms and background description is included of things like the malnutrition screening tools is included – perhaps as supplementary material.

- Author reply: We agree and have therefore included a 'Supplementary material 2' with an overview and characteristics of validated malnutrition screening tools (some of which are also included in the review). We have added to the discussion (line 304-305) that malnutrition assessment should always be preceded by screening with a reference to this Supplementary material 2.

- Revised section of the discussion: It should be stressed that we need to agree on using standardized methods for malnutrition assessment and interventions. However, malnutrition assessment should always be preceded by malnutrition screening with a validated tool (Supplementary material 2).

- Comment 3: The manuscript describes a systematic review of published papers in which malnutrition was assessed within populations of adult persons with (presumed or microbiologically confirmed) active TB disease and describes and evaluates the methods used for malnutrition assessment. The manuscript then attempts to determine if methods changed after the publication of the WHO guidelines on the management of malnutrition in TB in 2013.The stated aims, objectives, described methods and presented results are not fully consistent with each other and I suggest that the wording of these should be carefully reviewed and edited for clarity, precision and consistency. For example, the titland the primary aim at the end of the introduction state that the purpose is to describe current practice in TB, but the review is only of published research papers, which is unlikely to reflect general practice outside of research studies. I suggest changing the stated objectives to reflect this and/or including in the discussion.

- Author reply: We agree and have therefore now stated in the limitation section of the discussion that research and clinical practice are different settings and the results from our review are not an exact reflection of clinical practice (line 294-297). Nevertheless, we do expect BMI to be used most frequently in clinical practice, since the WHO recommends that BMI is the method to assess undernutrition/malnutrition. We have removed 'current/clinical practice' from the title and the rest of the manuscript, and we have carefully reviewed and edited for clarity, precision and consistency. More specifically, as the WHO (also in its guideline on nutritional care) recommends to use BMI as an assessment method we have removed the WHO publication from Table 3 as this publication therefore could not be expected to have changed the practice of using BMI (Table 3).

- Revised section: There are some limitations in our study. First, our study focused on published articles and not on the underlying study protocol. In some cases, detailed information on the assessment and operationalization of malnutrition might be available in the study protocol, however it was not addressed in the article making it unavailable to the public domain. Second, implementation of new criteria in in study protocols takes time. This implementation might take longer than the time period used in this review. Third: research and clinical practice are different settings and the results from our review are not a reflection of clinial practice. Nevertheless, we do expect BMI to be used most frequently in clinical practice since the WHO recommends that BMI is the method to assess undernutrition/malnutrition.

- Comment 4: The secondary aim/objective is to "quantify the malnutrition assessments that reflect the international consensus definition for malnutrition in this population" [intro, line 112-13] or "cover the three domains of the construct of malnutrition according to its international consensus definition" [abstract].

- Author reply: Thank you for pointing this out, we have changed in both places into: 'quantify how malnutrition assessment methods capture the international consensus definition for malnutrition'

- Comment 5: However, as stated by the authors [line 252, discussion] "the GLIM criteria do not include the domain of functionality as a criteria [sic]". Indeed, in the main, most recent GLIM document (Cederholm, Clinical Nutrition 2019) the document does not refer to these "domains" at all. Instead, the definitions are around definitions and diagnoses for phenotypic and aetiologic criteria for defining malnutrition. Further in the ESPEN guidelines on the definitions and terminology of clinical nutrition (ref 14) this uses these criteria to define into categories and sub-categories of disease-related malnutrition and malnutrition without disease, and these terms are not discussed in the current manuscript. I feel there is a real problem and confusion in the manuscript in trying to include physical and cognitive as both a conceptual domain of malnutrition and a diagnostic criterion. In disease-

related malnutrition it is impossible to separate out the impacts and aetiology of these from the disease or malnutrition.

- Author reply: Thank you for giving us the opportunity to clarify: our objective was to see if 'physical and cognitive function' (as part of the conceptual definition) was addressed by methods that are used to assess malnutrition, therefore it was categorized as domain C and not as a diagnostic criterion. To improve clarity on this, we have removed the ESPEN and GLIM criteria from Table 3, as well as the WHO guideline.

- Comment 6: Also, how do measures like grip strength (which was not included in the reported measures of malnutrition, but is included in some of the included manuscripts) fit into these domains. In the GLIM documents, grip-strength is included as a proxy for measuring body composition of reduced muscle mass (domain B), but yet is also clearly functional (domain C?).

- Author reply: Thank you for pointing this out. Handgrip strength is categorized in domain C and we have now added this in the manuscript line 196-197.

- Comment 7: The MNA that is include as an assessment tool for malnutrition is included as method that includes all three domains, but yet it is really a malnutrition screening tool and I question that it really addresses the "physical and cognitive domain" that people not familiar with the tool might conceive.

- Author reply: The MNA-SF is a screening tool and the MNA full is an assessment tool. For this review, we included both screening and assessment tools when the authors of the study both described and used it as an assessment tool.

- Comment 8: For the reporting of the results and characteristics of the included studies I suggest: (1) reporting and separating out studies based on why they assessed malnutrition, was it assessed as a primary outcome of the study, or as a main exposure or for other purposes and if so what? E.g. for determining those who were to receive some kind of intervention – nutritional or otherwise? (2) Related to this, reporting if participants in studies were inpatients or outpatients; (3) reporting and analysing studies by year of data collection rather than publication. In recent studies, this should always be reported.

- Author reply: We agree that this is important to report on and therefore we have added items 1, 2 and 3 to Table 4 and to the Results section. And also we do feel that this is an important direction for both future studies and clinical care and therefore we have added to the discussion that it is important that we agree on using standardized methods for screening, assessment and interventions (line 303-305).

- Revised section: Only two of the 17 different assessment methods that were used in the 69 studies captured all three domains of the definition, i.e., the Subjective Global Assessment (SGA) was used in two studies and the Mini Nutritional Assessment (MNA) was used in one study. Forty-one studies (59%) assessed malnutrition for the purpose of their primary aim. The three studies that used a method that captured all three domains of the definition, had a primary aim related to malnutrition. A descriptive subanalysis, as shown in Table 3, was performed to compare studies regarding their use of assessment tools, by their period of data collection. No differences were found between five-year periods regarding the use of BMI as the only assessment method.The studies that used methods that capture all three domains of the ESPEN definition of malnutrition did not describe when their data was collected Of the 69 included studies, 18 studies (26%) were on inpatients, 26 studies (38%) on outpatients, 7 (10%) on both type of patients and in 18 studies (26%) the type of patients was not described.

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- Comment 9: The objectives don't match the title of the manuscript.

- Author reply: The title has now been changed into: Malnutrition assessment methods in adult patients with tuberculosis - a systematic review

- Comment 10: The methods section needs more details of the search criteria. I suggest the whole abstract needs re-writing as per PRISMA guidelines and above comments

- Author reply: We have adapted the abstract in accordance with the PRISMA guidelines.

- Comment 11 (Introduction): The statement "70% of TB are malnourished needs referencing of the original source, not the Bhargava paper which concerns only a particular population in India.

- Author reply: We have replaced the reference for this statement with multiple references on the prevalence of malnutrition (line 105).

- Comment 12 (Methods): The methods state that the papers selected that "aimed to assess malnutrition and described a method for assessment of malnutrition in patients with microbiologically confirmed TB". (1) I know many of the studies included clinically diagnosed TB.

- Author reply: Thank you, we agree and changed this into 'with microbiologically confirmed or clinically diagnosed TB' (line 153-154).

Finally, please find included, as per your request, a revised document with and without track-changes.

Thank you again for considering our manuscript for publication.

VERSION 2 – REVIEW

REVIEWER	Cervantes, Jorge Texas Technical University Health Sciences Center - El Paso Campus
REVIEW RETURNED	10-Aug-2021

GENERAL COMMENTS	Since authors are not willing to make any further analysis of data that could be retrievable (as many of the parameters are in did measurable) and they just opted for using crosses (, +, ++), the soundness of the paper (i.e. just a recopilation of the assessment variables) does not bring anything truly valuable to the field.